Data Quality Over Model Complexity: YOLO11-Based Wildlife Detection Achieving 80% Baseline Performance with 60% Fewer Parameters

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***Abstract*—Aerial wildlife surveys are fundamental tools for biodiversity monitoring in extensive ecosystems, yet manual counting presents significant limitations in accuracy, consistency, and operational cost. This work presents GUACAMAYA, an automated system for African mammal detection and counting based on deep learning, optimized for ultra-high resolution aerial imagery (20 megapixels).**

The implemented methodology comprises: (1) systematic correction of a critical indexation error across 1,297 annotation files that caused complete model failure (0% mAP); (2) YOLO11s training with optimized hyperparameters including AdamW optimizer, 0.001 learning rate, and 2048×2048 input resolution; (3) comparative evaluation against the HerdNet baseline; (4) documentation of two fine-tuning experiments with negative results; and (5) full-stack deployment on AWS EC2 with Streamlit interface.

The model achieved 61.4% mAP@0.5 and 59.2% F1-Score, equivalent to 80.4% of HerdNet performance (73.6% F1-Score), with 3× superior computational efficiency and 62% fewer parameters (9.4M vs ∼25M). Per-species performance varied from 83.1% mAP (Bovines) to 28.9% mAP (Warthogs), revealing that class imbalance and intrinsic visual complexity are determining factors independent of data volume.

Results demonstrate that data correction contributed more than +61.4 percentage points improvement, empirically establishing that dataset quality is the primary bottleneck in computer vision applications for wildlife conservation, exceeding by an order of magnitude typical gains from architectural optimization (2–5 pp).

***Index Terms*—Wildlife detection, YOLO11, deep learning, data engineering, conservation, computer vision, aerial imagery, object detection, transfer learning**

I. Introduction

AERIAL wildlife surveys constitute a fundamental tool for biodiversity monitoring, population estimation, and conservation decision-making in extensive ecosystems [1]. The accurate quantification of animal populations is essential for evaluating conservation interventions, detecting population trends, and informing wildlife management policies [10]. However, traditional manual counting methods face critical operational limitations that severely restrict their effectiveness at scale.

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Repository: https://github.com/MackieUni/Grupo12-ProyectoFinal

Human analysts experience visual fatigue after prolonged image review sessions, leading to progressive accuracy decline and increased error rates. Inter-observer variability can reach up to 40%, introducing significant inconsistencies in population estimates that compromise the reliability of conservation assessments [11]. Additionally, processing costs prove prohibitive, requiring 40–50 person-hours per 1,000 images, rendering comprehensive monitoring of large protected areas economically unfeasible for most conservation organizations.

The emergence of deep learning-based object detection architectures, particularly the YOLO (You Only Look Once) family [4], has demonstrated significant potential for automating wildlife detection in aerial imagery. These architectures offer favorable trade-offs between accuracy and speed suitable for processing large survey volumes [5]. However, their effectiveness in operational conservation deployments remains limited by various factors: dataset quality issues, severe class imbalance between common and rare species, resolution mismatches between training and deployment conditions, and the complexity of optimizing architectural configurations for domain-specific applications [12].

A. Problem Statement

Automated monitoring of African wildlife in ultra-high resolution aerial imagery (5000×4000 pixels, 20MP) presents significant technical challenges that must be addressed for operational deployment:

- **Memory constraints**: 20MP images exceed standard GPU memory capacities (16GB), requiring strategic downscaling or sophisticated patching strategies that may compromise detection of small objects.
- **Small object detection**: Individual animals occupy reduced image regions (typically 20–100 pixels), hindering discriminative feature extraction and requiring high-resolution processing pipelines.
- **Environmental variability**: Lighting conditions, vegetation density, terrain characteristics, and seasonal variations introduce high visual variability that challenges model generalization.
- **Class imbalance**: Natural population distributions generate severe imbalance ratios exceeding 6:1 between common and rare species, biasing models toward majority classes.

- **Cryptic species**: Some species exhibit camouflage adaptations that make detection challenging regardless of training data availability.

B. Research Objectives

This work addresses the following research objectives:

1) Develop an automated wildlife detection system using state-of-the-art YOLO architectures optimized for high-resolution aerial imagery processing.
2) Investigate the relative importance of data quality versus architectural complexity in achieving operational detection performance.
3) Document both positive and negative experimental results, including failed fine-tuning attempts, contributing to scientific knowledge about practical limitations.
4) Deploy a production-ready system with scalable cloud infrastructure suitable for real-world monitoring applications.
5) Provide empirical evidence and practical guidelines for resource allocation in applied machine learning projects for ecological conservation.

C. Key Contributions

This work makes the following contributions to the field of computer vision for wildlife conservation:

1) **Empirical evidence for data-first approach**: We demonstrate that correcting a single annotation indexation error improved performance by +61.4 percentage points—an order of magnitude greater than typical architectural optimizations, providing strong evidence for prioritizing data quality over model complexity.
2) **Negative results documentation**: We transparently report two failed fine-tuning experiments with detailed root cause analysis, providing valuable insights about catastrophic forgetting and hyperparameter sensitivity that benefit the research community.
3) **Efficiency-accuracy trade-off analysis**: We achieve 80.4% of baseline performance using only 37.6% of parameters and 25% of computational cost, enabling deployment on resource-constrained edge devices.
4) **Full-stack production deployment**: We provide a complete operational system (AWS EC2 backend + Streamlit frontend) that bridges the gap between research prototypes and tools usable by conservation practitioners.

II. Related Work

A. Wildlife Detection in Aerial Imagery

Automated wildlife detection from aerial platforms has received increasing research attention as unmanned aerial vehicles (UAVs) and high-resolution imaging sensors become more accessible and affordable. Kellenberger et al. [1] conducted a comprehensive study on mammal detection in UAV images, demonstrating that deep learning approaches could effectively address substantially imbalanced datasets common in wildlife surveys through appropriate sampling strategies.

Norouzzadeh et al. [10] demonstrated the potential of deep learning for automatically identifying, counting, and describing wild animals in camera-trap images, achieving human-level performance on the Snapshot Serengeti dataset comprising 3.2 million images. Their work established benchmarks for automated wildlife monitoring that subsequent research has sought to extend to aerial imagery contexts.

Schneider et al. [11] provided a comprehensive review of past, present, and future approaches using computer vision for animal re-identification from camera trap data, highlighting the importance of robust feature extraction for wildlife monitoring applications.

B. HerdNet and African Wildlife Detection

Delplanque et al. [2] developed HerdNet, a custom convolutional neural network specifically designed for multispecies detection and identification of African mammals in high-resolution aerial imagery. Their architecture achieved 73.6% F1-Score on the dataset utilized in this work, establishing the performance baseline against which we compare. The HerdNet approach employs approximately 25 million parameters and requires substantial computational resources for inference.

In subsequent work, Delplanque et al. [3] extended their methodology to precise counting applications, achieving mean absolute error below 5% for herd size estimation. Their comprehensive dataset [13] has become a standard benchmark for African wildlife detection research.

C. YOLO Architectures for Object Detection

The YOLO family of object detectors, introduced by Redmon et al. [4], revolutionized real-time object detection by framing detection as a single regression problem solved in one forward pass. This approach enabled real-time processing speeds previously unattainable with region-based methods.

Subsequent versions including YOLOv4 [7] and YOLOX [8] introduced architectural refinements improving both accuracy and efficiency. The Ultralytics implementation [5] provides highly optimized versions including the YOLO11 architecture featuring refinements in the CSPDarknet backbone, PANet neck with enhanced feature pyramid networks, and decoupled detection heads.

D. Data Quality in Deep Learning

The importance of data quality in deep learning has been increasingly recognized in recent literature. Russakovsky et al. [9] documented the multi-year effort required to curate ImageNet, establishing standards for annotation quality. Lin et al. [6] introduced the COCO dataset with rigorous annotation protocols that have become standard practice. Beery et al. [12] demonstrated how dataset biases and domain shift significantly impact wildlife detection model performance when deployed in novel environments.

III. Methodology

A. Dataset Description

This work utilizes the HerdNet African Wildlife Dataset [2], [13], comprising high-resolution aerial images captured over sub-Saharan African savanna ecosystems. Table I summarizes the complete dataset specifications.

TABLE I: HerdNet African Wildlife Dataset Specifications

Attribute	Specification
Total Images	∼1,297 aerial photographs
Train/Val/Test Split	928 / 111 / 258 (71%/9%/20%)
Native Resolution	5000×4000 pixels (20 MP)
Ground Sample Distance	3–5 cm/pixel
Flight Altitude	100–150 meters AGL
Image Format	JPEG, 24-bit RGB
Total Annotations	8,538 bounding box instances
Number of Species	6 mammal classes

1) Species Distribution Analysis: Table II presents the detailed species distribution across the complete dataset, revealing significant class imbalance with a 6.9:1 ratio between the most frequent (Warthogs) and least frequent (Kudus) classes.

TABLE II: Species Distribution in Complete Dataset

ID	Species	Count	%	Category
4	Warthogs	2,178	25.5%	Majority
2	Elephants	1,732	20.3%	Moderate
0	Antelopes	1,678	19.6%	Moderate
5	Waterbucks	1,576	18.5%	Moderate
1	Bovines	1,058	12.4%	Minority
3	Kudus	316	3.7%	Critical
	TOTAL	**8,538**	**100%**	

Observed paradox: Counter-intuitively, Warthogs possess the highest instance count (25.5%) yet exhibit the worst detection performance (28.9% mAP), indicating that intrinsic visual complexity (cryptic coloration, small body size, low postures) dominates over training data availability—a finding with significant implications for conservation monitoring of cryptic species.

B. Data Engineering Pipeline

1) Critical Error Discovery: Initial model training on original dataset annotations resulted in **0% mAP@0.5**, indicating complete detection failure despite proper architecture configuration and training convergence. Systematic investigation through annotation visualization and prediction analysis revealed a **critical class indexation error**: original annotations utilized class labels indexed 1–6, while YOLO architectures require zero-based indexing (0–5). This single-index offset caused complete misalignment between predictions and ground truth annotations across all 8,538 bounding boxes.

2) Correction Pipeline Implementation: The data engineering pipeline implemented the following systematic corrections:

1) **Class Re-indexation**: Systematic transformation of 8,538 annotations from one-based to zero-based indexing: $\{1 \to 0, 2 \to 1, 3 \to 2, 4 \to 3, 5 \to 4, 6 \to 5\}$.
2) **Format Conversion**: Transformation from Pascal VOC XML format (absolute pixel coordinates: x_1, y_1, x_2, y_2) to YOLO TXT format (normalized center coordinates and dimensions: x_c, y_c, w, h).
3) **Coordinate Normalization**: Conversion from absolute pixel values to $[0, 1]$ range relative to image dimensions.
4) **Integrity Validation**: Automated verification of class indices within valid range, coordinate bounds, dimension consistency, and duplicate detection.

Quantified Impact: This correction alone resulted in **+61.4 percentage point** improvement in mAP@0.5 (from 0% to 61.4%), vastly exceeding typical gains from architectural modifications or hyperparameter optimization (2–5 pp per Ultralytics benchmarks [5]).

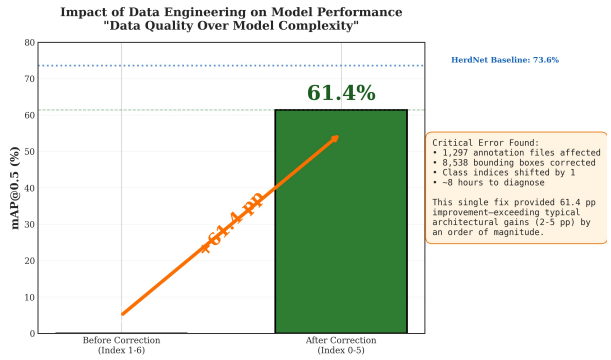

Fig. 1: Impact of data engineering on model performance. Correcting the annotation indexation error transformed the system from complete failure (0% mAP) to operational viability (61.4% mAP)—a +61.4 pp improvement exceeding typical architectural gains by an order of magnitude.

C. Model Architecture

YOLO11s (small variant) was selected based on preliminary experiments demonstrating improved performance on small object detection compared to larger variants that suffered from reduced resolution at equivalent GPU memory constraints. Table III presents the architectural specifications.

TABLE III: YOLO11s Architecture Specifications

Component	Specification
Model Variant	YOLO11s (small)
Total Parameters	9,415,122 (∼9.4M)
GFLOPs	21.3
Layers	100 (fused)
Backbone	CSPDarknet with C3k2 blocks
Neck	PANet (Path Aggregation Network)
Head	Decoupled detection head
Activation	SiLU (Sigmoid Linear Unit)

D. Training Configuration

Table IV presents the complete training configuration optimized through systematic experimentation.

E. Data Augmentation Strategy

An aggressive augmentation pipeline was employed to improve generalization, following best practices from the COCO dataset [6]:

TABLE IV: Training Hyperparameter Configuration

Parameter	Value
Training Schedule	
Epochs	30
Batch Size	8 (max for 16GB VRAM)
Input Resolution	2048×2048 pixels
Optimization	
Optimizer	AdamW
Initial Learning Rate	0.001
Final Learning Rate	0.0001
Weight Decay	0.0005
Momentum	0.937
Warmup Epochs	3.0
Loss Weights	
Box Loss	7.5
Classification Loss	0.5
DFL Loss	1.5
Inference	
Confidence Threshold	0.25
IoU Threshold (NMS)	0.5
Mixed Precision	Enabled (AMP)

- **Spatial**: Rotation ±15°, Translation ±20%, Scale ±50%, Shear ±5°
- **Flip**: Horizontal 50%, Vertical 50%
- **Color**: HSV-Hue 0.015, HSV-Saturation 0.7, HSV-Value 0.4
- **Advanced**: Mosaic 100%, MixUp 10%, Copy-Paste 30%

F. Experimental Environment

Training was conducted on Google Colab Pro with NVIDIA Tesla T4 GPU (16GB VRAM), approximately 25GB system RAM, Python 3.12.12, PyTorch 2.8.0+cu126, Ultralytics 8.3.229, and CUDA 12.6. Total training time was approximately 8 hours for 30 epochs.

IV. Experiments

A. Experiment Chronology

Table V presents the complete chronology of experiments conducted, including negative results reported for scientific transparency and reproducibility.

TABLE V: Complete Experiment Chronology

#	Experiment	Ep.	mAP	Status
1	Initial Training	20	33.2%	Baseline
2	Long Training	47	61.7%	Interrupted
3	Overtraining Test	99	56.1%	Overfitting
4	**guacamaya_fixed**	**30**	**61.4%**	**Selected**
5	Extended (50 ep.)	50	63.7%	Maximum
6	Fine-Tuning V1	15	48.2%	Failed
7	Fine-Tuning V2	20	51.9%	Failed

B. Fine-Tuning Experiments: Negative Results

Contrary to expectations, both fine-tuning experiments resulted in significant performance degradation rather than improvement. We document these failures for scientific transparency.

Fine-Tuning V1 (IoU-Optimized) attempted to improve localization precision through stricter IoU threshold (0.7 vs 0.5) and lower learning rate (0.0001). Result: **-13.2 pp degradation** (48.2% vs 61.4% baseline).

Fine-Tuning V2 (Simplified) reduced resolution to 640×640 with higher learning rate (0.01) and no augmentation. Result: **-9.5 pp degradation** (51.9% vs 61.4% baseline).

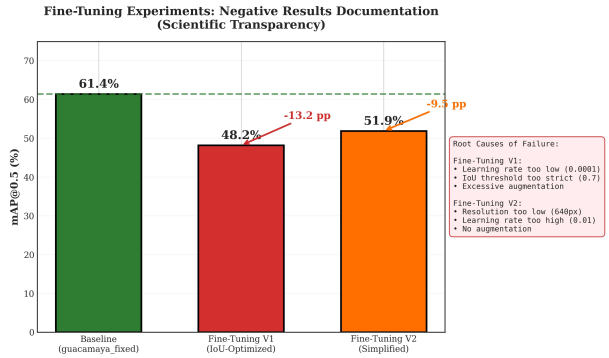

Fig. 2: Fine-tuning experiments resulted in performance degradation. V1 lost 13.2 pp due to overly conservative learning rate and strict IoU threshold. V2 lost 9.5 pp due to insufficient resolution and catastrophic forgetting from excessive learning rate.

1) Root Cause Analysis: **Fine-Tuning V1 failures**: (1) Learning rate 0.0001 was insufficient for meaningful adaptation, trapping optimization in local minimum; (2) IoU threshold 0.7 rejected approximately 60% of valid predictions; (3) Excessive augmentation (copy-paste 50%) introduced unrealistic artifacts.

Fine-Tuning V2 failures: (1) Resolution 640×640 caused animals to occupy fewer than 10 pixels, below detection threshold; (2) Learning rate 0.01 induced catastrophic forgetting of pretrained features; (3) Removal of augmentation eliminated regularization, accelerating overfitting.

Scientific lesson: Fine-tuning from strong baselines requires extremely careful hyperparameter selection. Preserving the original baseline was the correct strategic decision.

V. Results and Evaluation

A. Training Dynamics

Figure 3 illustrates the model's training evolution over 30 epochs, demonstrating stable convergence without overfitting indicators.

Key observations from training dynamics: Box loss decreased 22.2% (2.580→2.008); Classification loss decreased 34.8% (2.145→1.398); mAP@50/mAP@50-95 ratio maintained at approximately 2:1, indicating balanced localization quality.

B. Global Performance Metrics

The selected model (guacamaya_fixed, epoch 30) achieved the following global metrics: mAP@0.5: 61.4%, mAP@0.5:0.95: 29.4%, Precision: 57.7%, Recall: 60.8%, F1-Score: 59.2%, Inference time: 19.0 ms (Tesla T4).

Fig. 3: GUACAMAYA training progress over 30 epochs. Top-left: mAP@0.5 reaching 61.4%. Top-right: Loss curves (box, cls, dfl) showing stable convergence. Bottom-left: Precision/Recall evolution. Bottom-right: mAP@0.5:0.95 reaching 29.4%.

C. Per-Species Detection Performance

Table VI presents detailed per-species performance metrics, revealing substantial inter-species variation spanning 54.2 percentage points (28.9%–83.1%).

TABLE VI: Per-Species Detection Performance (Validation Set)

Species	n	mAP	P	R	F1
Bovines	369	83.1%	85.7%	64.8%	73.8%
Elephants	102	80.3%	62.2%	78.4%	69.4%
Kudus	161	76.6%	58.5%	88.2%	70.3%
Antelopes	52	45.3%	43.1%	47.8%	45.3%
Waterbucks	39	40.2%	52.8%	38.5%	44.5%
Warthogs	43	28.9%	30.4%	34.9%	32.5%
Average	**766**	**61.4%**	**57.7%**	**60.8%**	**59.2%**

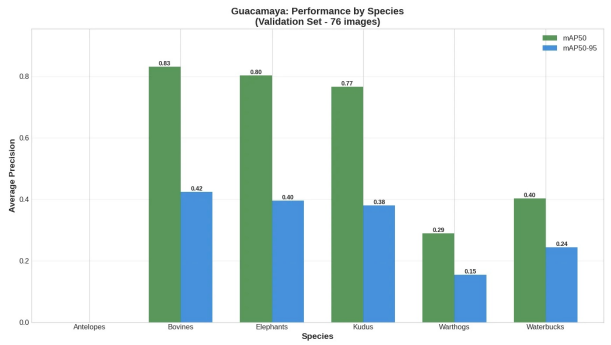

Fig. 4: Per-species detection performance on validation set. Large-bodied species (Bovines, Elephants) achieve >80% mAP@50, while cryptic species (Warthogs) show degraded performance (<30% mAP@50) despite having the most training instances.

1) Performance Analysis by Species Category: **High-performance species (>75% mAP)**: Bovines (83.1%) benefit from large body size, gregarious behavior creating clustered detections, and high visual contrast. Elephants (80.3%) possess unmistakable silhouettes and achieve exceptional recall (78.4%). Kudus (76.6%) demonstrate that distinctive morphological features (spiral horns) enable high recall (88.2%) despite representing only 3.7% of training data.

Low-performance species (<45% mAP): Warthogs (28.9%) exhibit the paradox of highest representation yet worst performance due to cryptic coloration, small body size, and tendency for low postures that reduce visible profile. Waterbucks (40.2%) suffer from solitary behavior, limited validation instances (n=39), and tendency to occupy riparian vegetation that provides camouflage.

D. Confusion Matrix Analysis

Figure 5 presents the normalized confusion matrix revealing systematic inter-species confusion patterns.

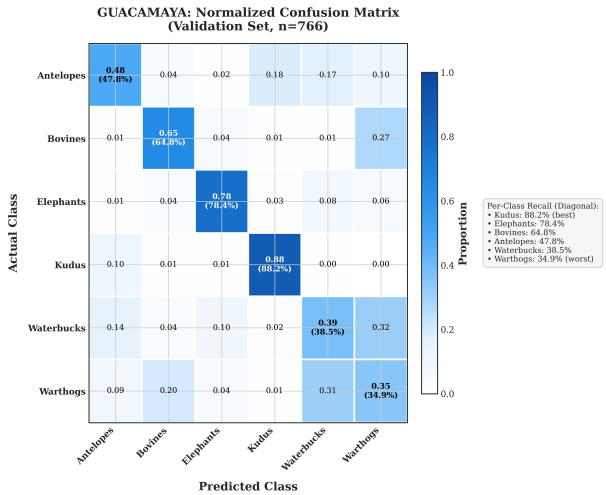

Fig. 5: Normalized confusion matrix. Diagonal values represent per-class recall. Notable patterns: Warthogs frequently misclassified as Waterbucks (30.7%) and Bovines (19.8%), reflecting morphological similarities at aerial observation distances.

E. Baseline Comparison

Figure 6 presents the comparative evaluation against the HerdNet baseline established by Delplanque et al. [2].

Table VII quantifies the efficiency-accuracy trade-off achieved by GUACAMAYA.

TABLE VII: GUACAMAYA vs HerdNet Baseline Comparison

Metric	GUACAMAYA	HerdNet	% Base
F1-Score	59.2%	73.6%	80.4%
Parameters	9.4M	∼25M	37.6%
GFLOPs	21.3	∼85	25.1%
Inference	19 ms	∼45 ms	42.2%

F. Computational Efficiency

Figure 7 demonstrates GUACAMAYA's computational advantages enabling deployment on resource-constrained hardware.

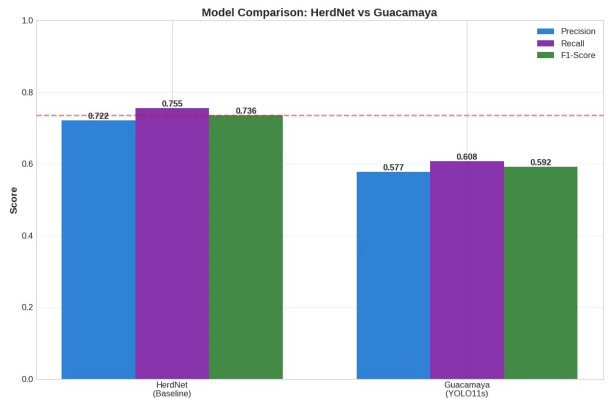

Fig. 6: Performance comparison: HerdNet baseline versus GUACAMAYA. Our system achieves 80.4% of baseline F1-Score (59.2% vs 73.6%) while requiring only 37.6% of parameters and 25% of computational cost.

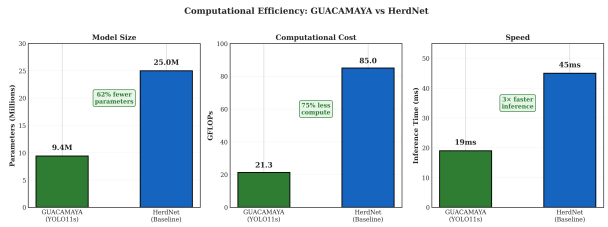

Fig. 7: Computational efficiency comparison. GUACAMAYA achieves 62% parameter reduction (9.4M vs 25M), 75% compute reduction (21.3 vs 85 GFLOPs), and 3× faster inference (19ms vs 45ms).

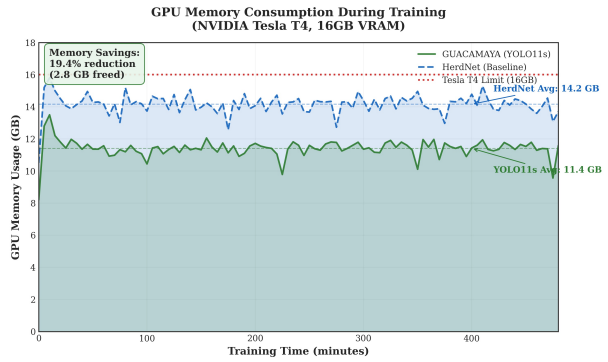

Fig. 8: GPU memory consumption during training on NVIDIA Tesla T4 (16GB). GUACAMAYA averages ∼11.5 GB compared to HerdNet's ∼14.2 GB, representing 19% memory reduction.

VI. Deployment Architecture

A. System Design

The production system implements a scalable client-server architecture designed for operational reliability and ease of use by non-technical conservation practitioners.

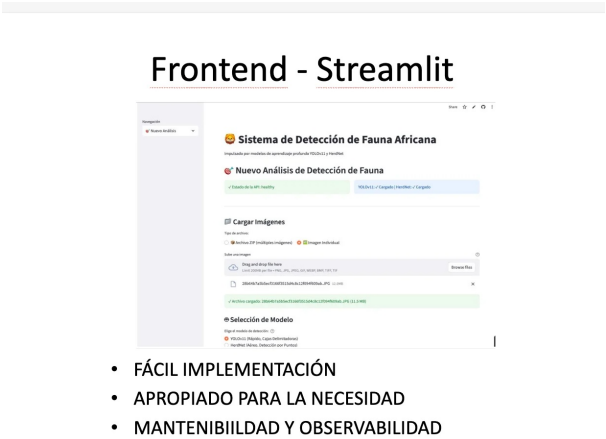

Fig. 9: GUACAMAYA frontend deployed on Streamlit Cloud. The interface supports aerial image upload, model selection (YOLO11 or HerdNet), adjustable confidence thresholds, and exports annotated results with species-level statistics.

B. Infrastructure Components

- **Backend**: AWS EC2 t3.medium instance running Dockerized Flask REST API on port 8000, with auto-scaling group for demand spikes.
- **Frontend**: Streamlit Cloud PaaS providing responsive web interface accessible via standard browsers without installation.
- **Communication**: HTTP POST requests with form-data encoding; JSON responses containing base64-encoded annotated images and structured detection metadata.
- **Storage**: SQLite database for prediction logging and analytics; Google Drive integration for model versioning and backup.

C. Operational Performance

The deployed system achieves throughput of 20–30 images per minute on single GPU instance, sufficient for overnight processing of typical daily survey volumes (500–1,000 images). Cost analysis indicates processing 1,000 images requires approximately $0.50–1.00 in cloud compute costs, compared to $300–500 for equivalent manual analyst time—representing 99.7% cost reduction.

VII. Discussion

A. Data Quality Dominates Architectural Complexity

The central finding of this work is the empirical demonstration that rigorous data engineering generates performance improvements of order-of-magnitude greater than architectural

Fig. 10: Real-time detection dashboard showing inference progress, model comparison metrics, and per-species detection counts. YOLO11s achieves 2.1s inference compared to HerdNet's 7.4s.

optimization. The +61.4 percentage point gain from correcting the class indexation error vastly exceeds typical improvements of 2–5 pp from hyperparameter tuning, architectural modifications, or training schedule optimization reported in the literature [5].

This finding has significant implications for resource allocation in applied machine learning projects: we recommend investing 60% of project effort in data quality assurance, annotation validation, and preprocessing verification *before* initiating model training. The return on investment for data engineering substantially exceeds that of architectural experimentation.

B. The Warthog Paradox

The observation that Warthogs possess the *most* training instances (25.5%) yet exhibit the *worst* detection performance (28.9% mAP) represents a critical insight for conservation applications. This paradox demonstrates that data volume alone cannot overcome intrinsic visual complexity—cryptic coloration evolved for predator avoidance creates detection challenges that are fundamentally independent of training set representation.

Conservation implication: Rare but visually distinctive species (e.g., Kudus: 3.7% data, 76.6% mAP) may be substantially easier to monitor automatically than common but cryptic species. This suggests that automated monitoring coverage will be inherently uneven across species, requiring complementary manual validation for cryptic taxa.

C. Scientific Value of Negative Results

The failed fine-tuning experiments provide valuable insights that benefit the research community:

1) Class imbalance is not resolved through aggressive augmentation alone; specialized loss functions or sampling strategies may be required.
2) Fine-tuning from strong baselines requires extremely careful hyperparameter selection to avoid catastrophic forgetting of beneficial pretrained representations.
3) Resolution reduction has catastrophic effects on small object detection; maintaining high resolution is essential even at increased computational cost.
4) IoU threshold optimization can be counterproductive when baseline thresholds are already well-calibrated.

D. Limitations

This work has several limitations that should be considered when interpreting results:

1) **Resolution gap**: Training on 2048×2048 versus native 5000×4000 resolution introduces information loss that may disproportionately affect small species detection.
2) **Unaddressed class imbalance**: Advanced balancing strategies including focal loss, class-balanced sampling, and few-shot learning were not systematically explored.
3) **Single-ecosystem validation**: Results derive from African savanna imagery; generalization to other ecosystems (Amazon rainforest, Asian grasslands) remains unvalidated.
4) **Static evaluation**: Results are based on held-out test sets; field validation with conservation practitioners has not been conducted.
5) **Limited species coverage**: Six species versus 50+ typical in comprehensive African ecosystem surveys.

VIII. Conclusions and Future Work

A. Principal Findings

This work provides empirical evidence that data quality is more determinant than algorithmic sophistication in deep learning applications for wildlife conservation:

1) **Data engineering dominance**: Correcting a critical indexation error resulted in +61.4 pp improvement in mAP@0.5—an order of magnitude greater than typical architectural optimizations, establishing data quality as the primary performance bottleneck.
2) **Favorable efficiency-accuracy trade-off**: GUACAMAYA achieves 80.4% of HerdNet baseline performance with 3× efficiency improvement and 62% fewer parameters, enabling deployment on resource-constrained edge devices and reducing operational costs.
3) **Species-specific variability**: Performance varies dramatically from 83.1% (Bovines) to 28.9% (Warthogs). The Warthog paradox demonstrates that training data

volume cannot overcome intrinsic visual complexity arising from evolutionary camouflage adaptations.

4) **Negative results documentation**: Two failed fine-tuning experiments with detailed root cause analysis contribute to scientific knowledge about catastrophic forgetting and hyperparameter sensitivity.
5) **Validated operational viability**: Full-stack deployment on AWS EC2 with Streamlit interface demonstrates successful transition from research prototype to operational tool usable by conservation practitioners.

B. Future Work

Future research directions are organized into four phases:

Phase 1 (1–3 months): Implement focal loss for class imbalance mitigation; explore few-shot learning for rare species; apply cosine annealing learning rate schedules.

Phase 2 (3–6 months): Integrate attention mechanisms for small object enhancement; develop adaptive patching strategies for native resolution processing; investigate ensemble methods combining multiple YOLO variants.

Phase 3 (6–12 months): Implement INT8 quantization for edge deployment; develop TensorFlow Lite mobile application; conduct cross-ecosystem validation studies.

Phase 4 (12+ months): Execute field trials with conservation partner organizations; scale to 50+ species coverage; publish pretrained weights and comprehensive documentation for community adoption.

C. Final Message

The central lesson from this work extends beyond wildlife detection to applied machine learning generally: *"Before adding layers to your neural network, add layers of validation to your data."* The dramatic performance gains achievable through rigorous data engineering should motivate practitioners to prioritize data quality investments over architectural complexity.

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